

Task 2 SQA

Sample E-commerce Shopping System – QA Test Document

1. Test Scenarios

Scenario ID	Module	Test Scenario	Testing Type
TS_001	Registration	Verify user can register with valid details.	Positive
TS_002	Registration	Verify registration fails with an invalid email format.	Negative
TS_003	Registration	Verify registration fails when password contains fewer than 8 characters.	Boundary/Negative
TS_004	Registration	Verify registration fails when required fields are left empty.	Negative
TS_005	Registration	Verify registration succeeds with a password of exactly 8 characters.	Boundary/Positive
TS_006	Login	Verify user can log in with valid email and password.	Positive
TS_007	Login	Verify an error message is displayed for an incorrect password.	Negative
TS_008	Login	Verify an error message is displayed for an unregistered email address.	Negative
TS_009	Login	Verify login is blocked when email and password fields are empty.	Negative
TS_010	Product Search	Verify search returns matching products for a valid keyword.	Positive
TS_011	Product Search	Verify an appropriate message is displayed when no matching products are found.	Negative
TS_012	Product Search	Verify search behavior when the search field is left empty.	Negative
TS_013	Shopping Cart	Verify a product can be added to the shopping cart.	Positive
TS_014	Shopping Cart	Verify product quantity can be increased and decreased.	Positive
TS_015	Shopping Cart	Verify a product can be removed from the shopping cart.	Positive
TS_016	Shopping Cart	Verify the cart total updates correctly after quantity changes.	Functional
TS_017	Checkout	Verify an order is placed successfully with valid delivery information and a selected payment method.	Positive
TS_018	Checkout	Verify an order is not placed when required delivery information is missing.	Negative
TS_019	Checkout	Verify an order is not placed when no payment method is selected.	Negative
TS_020	Checkout	Verify an appropriate confirmation message is displayed after successful order placement.	Functional

2. Test Cases (TC_001 – TC_025)

Test Case ID	TC_001
Module	Registration
Test Scenario	Verify user can register with valid details
Precondition	User is on the Registration page
Test Steps	1. Enter a valid name. 2. Enter a valid email address. 3. Enter a password with 8 or more characters. 4. Click Register.
Test Data	Name: Ali Khan Email: ali@test.com Password: Test@123
Expected Result	User account is created successfully.
Actual Result	Not Executed
Status	Not Executed
Test Case ID	TC_002
Module	Registration
Test Scenario	Verify registration fails with an invalid email format
Precondition	User is on the Registration page
Test Steps	1. Enter valid name. 2. Enter an invalid email (alitest.com). 3. Enter a valid password. 4. Click Register.
Test Data	Email: alitest.com
Expected Result	System should display an "Invalid email format" message and prevent registration.
Actual Result	Not Executed
Status	Not Executed
Test Case ID	TC_003
Module	Registration
Test Scenario	Verify registration fails with a password shorter than 8 characters
Precondition	User is on the Registration page
Test Steps	1. Enter valid name and email. 2. Enter a 7-character password. 3. Click Register.
Test Data	Password: Test123 (7 characters)
Expected Result	Registration should fail with a password validation message.
Actual Result	Not Executed

Status	Not Executed
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Test Case ID	TC_004
Module	Registration
Test Scenario	Verify registration fails when required fields are empty
Precondition	User is on the Registration page
Test Steps	1. Leave all required fields empty. 2. Click Register.
Test Data	Blank fields
Expected Result	System should display validation messages for all required fields.
Actual Result	Not Executed
Status	Not Executed
Test Case ID	TC_005
Module	Registration
Test Scenario	Verify registration succeeds with exactly 8-character password
Precondition	User is on the Registration page
Test Steps	1. Enter valid details. 2. Enter an 8-character password. 3. Click Register.
Test Data	Password: Test@123
Expected Result	Registration should be successful.
Actual Result	Not Executed
Status	Not Executed
Test Case ID	TC_006
Module	Login
Test Scenario	Verify login with valid credentials
Precondition	A registered user account exists.
Test Steps	1. Open the Login page. 2. Enter a valid email and password. 3. Click Login.
Test Data	Email: ali@test.com Password: Test@123
Expected Result	User is logged in successfully.
Actual Result	Not Executed
Status	Not Executed

Test Case ID	TC_007
Module	Login
Test Scenario	Verify login fails with an incorrect password
Precondition	A registered user account exists.
Test Steps	1. Enter a valid email. 2. Enter an incorrect password. 3. Click Login.
Test Data	Email: ali@test.com Password: Wrong@123
Expected Result	System should display an "Invalid credentials" message and deny access.
Actual Result	Not Executed
Status	Not Executed

Test Case ID	TC_008
Module	Login
Test Scenario	Verify login fails with an unregistered email
Precondition	User is on the Login page.
Test Steps	1. Enter an unregistered email. 2. Enter any valid password. 3. Click Login.
Test Data	Email: unknown@test.com Password: Test@123
Expected Result	System should display an "Invalid credentials" message.
Actual Result	Not Executed
Status	Not Executed
Test Case ID	TC_009
Module	Login
Test Scenario	Verify login is blocked when fields are empty
Precondition	User is on the Login page.
Test Steps	1. Leave email and password blank. 2. Click Login.
Test Data	Blank email and password
Expected Result	System should not log in and should display validation messages.
Actual Result	Not Executed
Status	Not Executed
Test Case ID	TC_010

Module	Product Search
Test Scenario	Verify search returns matching products
Precondition	Products are available in the catalog.
Test Steps	1. Enter "Laptop" in the search bar. 2. Click Search.
Test Data	Keyword: Laptop
Expected Result	Matching products related to "Laptop" should be displayed.
Actual Result	Not Executed
Status	Not Executed
Test Case ID	TC_011
Module	Product Search
Test Scenario	Verify appropriate message is displayed when no product is found
Precondition	User is on the Product Search page
Test Steps	1. Enter a keyword that does not exist (e.g., "XYZ123"). 2. Click Search.
Test Data	XYZ123
Expected Result	System should display an appropriate message such as "No products found."
Actual Result	Not Executed
Status	Not Executed

Test Case ID	TC_012
Module	Product Search
Test Scenario	Verify search behavior when the search field is empty
Precondition	User is on the Product Search page
Test Steps	1. Leave the search field blank. 2. Click Search.
Test Data	Blank
Expected Result	System should display a validation message or show all products, depending on system design.
Actual Result	Not Executed
Status	Not Executed
Test Case ID	TC_013
Module	Shopping Cart
Test Scenario	Verify user can add a product to the cart
Precondition	User is viewing a product

Test Steps	1. Select a product. 2. Click Add to Cart.
Test Data	Product: Laptop
Expected Result	Product should be added to the shopping cart successfully.
Actual Result	Not Executed
Status	Not Executed
Test Case ID	TC_014
Module	Shopping Cart
Test Scenario	Verify user can increase product quantity
Precondition	Product already exists in the cart
Test Steps	1. Open the cart. 2. Increase product quantity from 1 to 2.
Test Data	Quantity = 2
Expected Result	Product quantity and cart total should update correctly.
Actual Result	Not Executed
Status	Not Executed
Test Case ID	TC_015
Module	Shopping Cart
Test Scenario	Verify user can decrease product quantity
Precondition	Product quantity is greater than 1
Test Steps	1. Open the cart. 2. Decrease quantity by one.
Test Data	Quantity: 2 → 1
Expected Result	Quantity and total price should update correctly.
Actual Result	Not Executed
Status	Not Executed

Test Case ID	TC_016
Module	Shopping Cart
Test Scenario	Verify user can remove a product from the cart
Precondition	Cart contains at least one product
Test Steps	1. Open the cart. 2. Click Remove for a product.
Test Data	Product: Laptop

Expected Result	Selected product should be removed and cart total updated.
Actual Result	Not Executed
Status	Not Executed
Test Case ID	TC_017
Module	Shopping Cart
Test Scenario	Verify cart total updates after increasing quantity
Precondition	Product exists in cart
Test Steps	1. Increase quantity. 2. Observe the total amount.
Test Data	Product Price: \$100, Quantity: 2
Expected Result	Total should update to \$200.
Actual Result	Not Executed
Status	Not Executed
Test Case ID	TC_018
Module	Checkout
Test Scenario	Verify successful checkout with valid delivery information
Precondition	Cart contains at least one product
Test Steps	1. Proceed to Checkout. 2. Enter all required delivery details. 3. Select a payment method. 4. Click Place Order.
Test Data	Valid Name, Address, Phone, Payment Method
Expected Result	Order should be placed successfully.
Actual Result	Not Executed
Status	Not Executed
Test Case ID	TC_019
Module	Checkout
Test Scenario	Verify order is not placed when delivery information is missing
Precondition	Product exists in cart
Test Steps	1. Go to Checkout. 2. Leave required delivery fields empty. 3. Click Place Order.
Test Data	Missing Address
Expected Result	System should prevent order placement and display validation messages.
Actual Result	Not Executed

Status	Not Executed
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Test Case ID	TC_020
Module	Checkout
Test Scenario	Verify order is not placed without selecting a payment method
Precondition	Product exists in cart
Test Steps	1. Fill all delivery information. 2. Do not select any payment method. 3. Click Place Order.
Test Data	Payment Method: None
Expected Result	Order should not be placed and user should be prompted to select a payment method.
Actual Result	Not Executed
Status	Not Executed
Test Case ID	TC_021
Module	Registration
Test Scenario	Verify registration fails with a 7-character password (Boundary Value)
Precondition	User is on Registration page
Test Steps	Enter valid details with a 7-character password and click Register.
Test Data	Password: Abc1234
Expected Result	Registration should fail.
Actual Result	Not Executed
Status	Not Executed
Test Case ID	TC_022
Module	Registration
Test Scenario	Verify registration succeeds with an 8-character password (Boundary Value)
Precondition	User is on Registration page
Test Steps	Enter valid details with an 8-character password and click Register.
Test Data	Password: Abc12345
Expected Result	Registration should succeed.
Actual Result	Not Executed
Status	Not Executed
Test Case ID	TC_023
Module	Registration

Test Scenario	Verify registration succeeds with a 9-character password (Boundary Value)
Precondition	User is on Registration page
Test Steps	Enter valid details with a 9-character password and click Register.
Test Data	Password: Abc123456
Expected Result	Registration should succeed.
Actual Result	Not Executed
Status	Not Executed

Test Case ID	TC_024
Module	Login
Test Scenario	Verify login is blocked when only the email is entered
Precondition	User is on Login page
Test Steps	1. Enter a valid email. 2. Leave password empty. 3. Click Login.
Test Data	Email: ali@test.com
Expected Result	System should display a validation message and prevent login.
Actual Result	Not Executed
Status	Not Executed
Test Case ID	TC_025
Module	Login
Test Scenario	Verify login is blocked when only the password is entered
Precondition	User is on Login page
Test Steps	1. Leave email empty. 2. Enter a valid password. 3. Click Login.
Test Data	Password: Test@123
Expected Result	System should display a validation message and prevent login.
Actual Result	Not Executed
Status	Not Executed

4. Positive & Negative Test Cases

Positive Test Cases

Test Case ID	Module	Test Description	Expected Result
TC_001	Registration	Register with valid details	Account is created successfully
TC_005	Registration	Register using an 8-character password	Registration succeeds
TC_006	Login	Login using valid email and password	User logs in successfully
TC_010	Product Search	Search using a valid product keyword	Matching products are displayed
TC_013	Shopping Cart	Add a product to the cart	Product is added successfully
TC_014	Shopping Cart	Increase product quantity	Quantity and total are updated
TC_015	Shopping Cart	Decrease product quantity	Quantity and total are updated
TC_016	Shopping Cart	Remove a product from the cart	Product is removed successfully
TC_018	Checkout	Checkout with valid delivery information and payment method	Order is placed successfully

Negative Test Cases

Test Case ID	Module	Test Description	Expected Result
TC_002	Registration	Register using an invalid email	Registration is blocked and an error is displayed
TC_003	Registration	Register with a password shorter than 8 characters	Registration is blocked
TC_004	Registration	Leave required registration fields empty	Validation messages are displayed
TC_007	Login	Login using an incorrect password	Invalid credentials message is displayed
TC_008	Login	Login using an unregistered email	Invalid credentials message is displayed
TC_009	Login	Leave email and password empty	Login is not allowed
TC_011	Product Search	Search for a non-existing product	"No products found" message is displayed
TC_019	Checkout	Leave required delivery information empty	Order is not placed
TC_020	Checkout	Do not select a payment method	Order is not placed
TC_024	Login	Enter email only	Login is blocked
TC_025	Login	Enter password only	Login is blocked

5. Boundary Value Testing

Requirement: Password must contain at least 8 characters.

Test Case ID	Input	Expected Result	Result
BVA_001	7-character password (Abc1234)	Registration should fail	Pass if rejected
BVA_002	8-character password (Abc12345)	Registration should succeed	Pass if accepted
BVA_003	9-character password (Abc123456)	Registration should succeed	Pass if accepted

Reason: These values test the boundary around the minimum password length requirement.

6. Bug Reports (Hypothetical)

Since the website is not available, these are hypothetical bugs based strictly on the requirements.

Bug Report 1

Bug ID	BUG_001
Bug Title	Registration accepts a password with fewer than 8 characters
Module	Registration
Steps to Reproduce	1. Open Registration page. 2. Enter valid name and email. 3. Enter a 7-character password. 4. Click Register.
Expected Result	Registration should fail with a validation message.
Actual Result	Registration is completed successfully.
Severity	High
Priority	High

Bug Report 2

Bug ID	BUG_002
Bug Title	Login succeeds with an incorrect password
Module	Login
Steps to Reproduce	1. Enter a valid email. 2. Enter an incorrect password. 3. Click Login.
Expected Result	System should display an "Invalid credentials" message.
Actual Result	User is logged in successfully.
Severity	Critical
Priority	High

Bug Report 3

Bug ID	BUG_003
Bug Title	Shopping cart total does not update after increasing product quantity
Module	Shopping Cart
Steps to Reproduce	1. Add a product to the cart. 2. Increase the quantity. 3. Observe the total amount.
Expected Result	Cart total should update according to the new quantity.
Actual Result	Cart total remains unchanged.
Severity	High
Priority	Medium

Bug Report 4

Bug ID	BUG_004
Bug Title	Order is placed without required delivery information
Module	Checkout
Steps to Reproduce	1. Add a product to the cart. 2. Go to Checkout. 3. Leave the delivery address empty. 4. Click Place Order.
Expected Result	System should prevent order placement and display validation messages.
Actual Result	Order is placed successfully.
Severity	Critical
Priority	High

Bug Report 5

Bug ID	BUG_005
Bug Title	No message is displayed when no products match the search keyword
Module	Product Search
Steps to Reproduce	1. Enter a keyword that does not match any product. 2. Click Search.
Expected Result	System should display an appropriate "No products found" message.
Actual Result	The page remains blank with no message.
Severity	Medium
Priority	Medium